

Placement Portal Application — Project Report

1. Student Details

Name: Aditya Kashyap Mohanty

Roll Number: 24f3005137

Email: 24f3005137@ds.study.iitm.ac.in

About Me:

I am a student in the IIT Madras BS Degree program with a strong interest in applying machine learning to embedded systems. I enjoy building structured and scalable applications that solve real-world workflow problems.

2. Project Details

Project Title: Placement Portal Application

Problem Statement:

The placement process in many institutions lacks a centralized system to manage student applications, company recruitment drives, and administrative approvals. This project aims to develop a web-based platform that streamlines interactions between students, companies, and administrators, ensuring transparency, efficiency, and structured data management.

Approach:

The application is built using Flask with a modular architecture based on Blueprints. It implements a role-based system for Admin, Student, and Company users. Students can apply to approved placement drives, companies can manage recruitment, and admins oversee approvals. The system includes resume uploads, application tracking, and validation mechanisms to ensure data integrity.

3. AI/LLM Declaration

I used ChatGPT for assistance in debugging, structuring certain parts of the code, and improving naming conventions.

The extent of AI usage is approximately 15%, limited to guidance and suggestions.

All final implementation, logic building, and integration were done independently.

4. Technologies and Frameworks Used

Technology	Purpose
Flask	Backend web framework
SQLAlchemy	ORM for database interactions
SQLite	Lightweight relational database
Jinja2	Template engine for dynamic HTML rendering
Bootstrap 5	Frontend styling and responsiveness
Flask-Login	Authentication and session management
Werkzeug	Password hashing and file handling

5. Database Schema / Design

User

- id, username, email, password (hashed), role, is_active

StudentProfile

- user_id, name, contact, cgpa, department, resume_path

CompanyProfile

- user_id, name, hr_contact, website, description, approval_status

PlacementDrive

- company_id, job_title, job_description, eligibility_criteria, deadline, status

Application

- student_id, drive_id, application_date, status, remarks

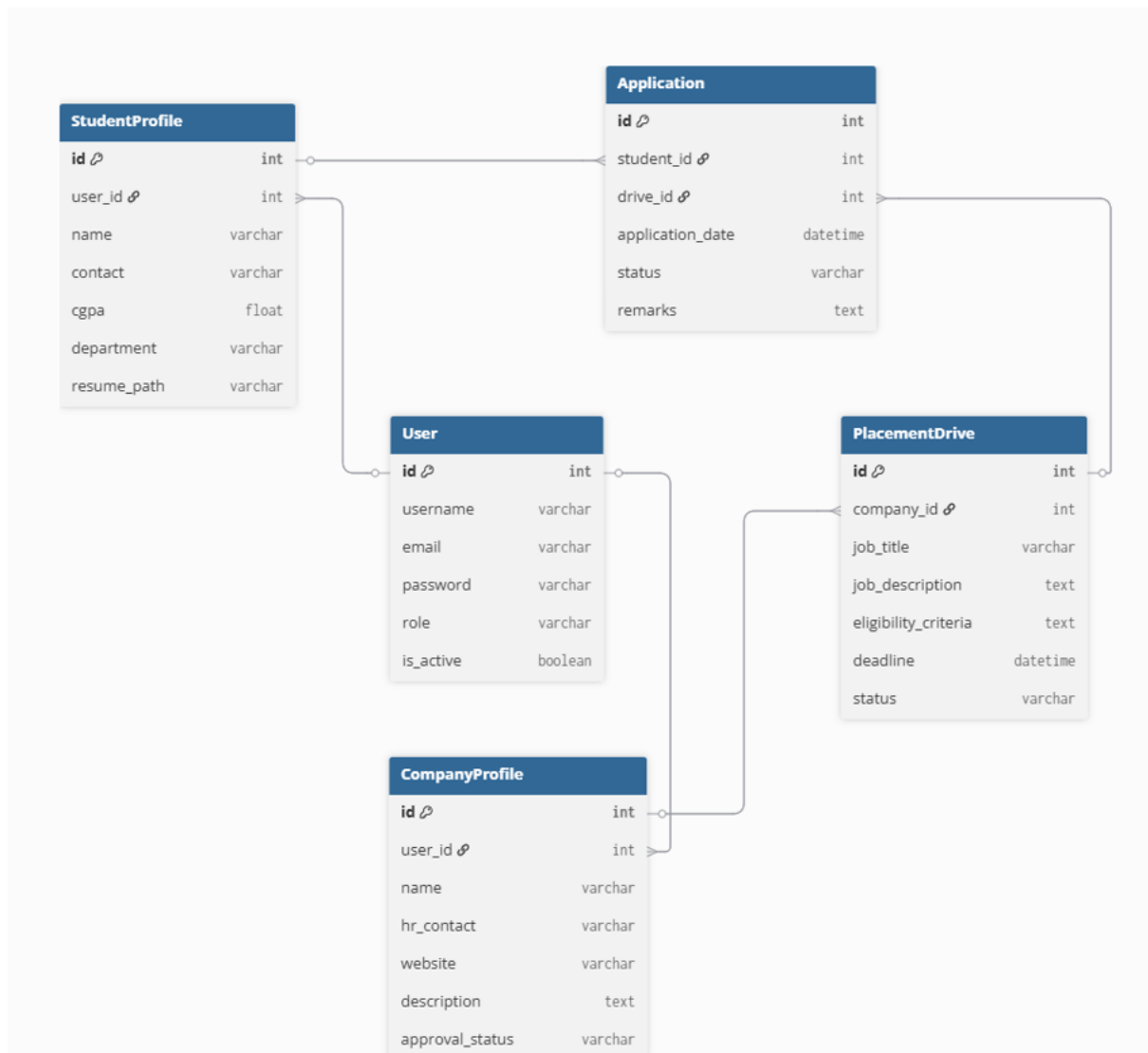
Relationships:

- One-to-Many: User → StudentProfile / CompanyProfile
- One-to-Many: CompanyProfile → PlacementDrive
- One-to-Many: StudentProfile → Application
- One-to-Many: PlacementDrive → Application

Constraint:

- UniqueConstraint(student_id, drive_id) ensures no duplicate applications

ER – Diagram



The lines in the diagram denote the connections between different tables through the foreign keys.

6. API Resource Endpoints

Endpoint	Method	Description
/api/companies	GET	Fetch all approved companies
/api/drives	GET	Fetch all approved placement drives

A YAML API definition file (api.yaml) is included in the submission.

7. Architecture and Features

Architecture Overview:

- **app.py** → Entry point and app initialization
 - **models.py** → Database models
 - **routes/** → Modular route handling (auth, admin, student, company)
 - **templates/** → HTML templates using Jinja2
 - **static/** → CSS and uploaded files
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Core Features Implemented:

- Role-based authentication (Admin, Student, Company)
- Admin approval system for companies and drives
- Placement drive creation and management
- Student application system with validation
- Resume upload with PDF validation
- Application tracking with status updates
- Duplicate application prevention

- Deadline and eligibility checks
 - CSV export of applicants (Company side)
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Additional Features:

- **Recommendation System:**
Drives are highlighted based on student CGPA and department using pattern matching logic.
 - **Search and Filtering:**
Students can search for drives using keywords.
 - **Dashboard Analytics:**
Admin dashboard includes system statistics using charts.
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8. Video Presentation

Drive Link:

[MAD1 – 24f3005137 -PLACEMENT PORTAL](#)

9. Conclusion

The Placement Portal successfully implements a complete recruitment workflow system with structured role-based access, validations, and user-friendly interfaces. The project demonstrates strong backend logic, database design, and practical application of web development concepts. Additional features such as recommendation logic and CSV export enhance the usability and completeness of the system.
